

Yajie Zhou

Nanchang, Jiangxi, China

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EDUCATION

Nanchang University

Master of Medicine (Surgery) | Weighted Average: 84.27/100

Nanchang, China

Sep 2024–Present

Jinggangshan University

Bachelor of Medicine in Clinical Medicine | Weighted Average: 82.35/100

Ji'an, China

Sep 2019–Jun 2024

RESEARCH PROFILE

Clinically trained cancer bioinformatics researcher with first-author experience in formulating questions from literature gaps, designing end-to-end computational studies, and translating multi-omics analyses into interpretable figures and manuscripts. My goal is to advance toward mechanistically grounded single-cell and spatial oncology research.

PUBLICATIONS

Peer-reviewed First-Author Publications

- **Zhou, Y.**, Hu, Z., Xie, L., Zhang, W., & Rao, H. (2026). Comprehensive multi-omics profiling of tertiary lymphoid structures reveals immunogenetic landscapes and prognostic subtypes in lung adenocarcinoma. *Molecular Carcinogenesis*, 65(7), 888–903. doi:10.1002/mc.70120
- **Zhou, Y.**, Hu, Z., Xie, L., Zhang, S., Huang, Q., Zhang, N., Liu, Y., Zhang, W., & Wang, K. (2026). Identification and external validation of a prognostic signature based on bone morphogenetic protein-related mRNAs for kidney renal clear cell carcinoma. *Discover Oncology*. doi:10.1007/s12672-026-05238-z
- Xie, L., **Zhou, Y.**, Hu, Z., Zhang, W., & Zhang, X. (2025). Integrative multi-omics reveals energy metabolism-related prognostic signatures and immunogenetic landscapes in lung adenocarcinoma. *Frontiers in Immunology*, 16, 1679464. doi:10.3389/fimmu.2025.1679464

Manuscript Under Review

- **Zhou, Y.**, Hu, Z., Xie, L., Lai, Y., Kuang, J., Zhang, W., & Xie, L. Multi-omics characterization of ENO1-associated mitophagy–metabolism crosstalk and tumor microenvironment remodeling in lung adenocarcinoma. *Under review*.

RESEARCH EXPERIENCE

ENO1-associated Mitophagy–Metabolism Crosstalk and Tumor Microenvironment Remodeling in Lung Adenocarcinoma

Nanchang, China

Graduate Researcher and First Author, Nanchang University

2025–Present

- Formulated the central question from literature gaps and designed an integrative framework to study ENO1-associated metabolic epithelial states and spatial tumor niches in lung adenocarcinoma.
- Independently learned and integrated bulk RNA-seq, survival modeling, WGCNA, single-cell RNA-seq, spatial transcriptomics, pseudotime analysis, and CellChat-based communication inference; resolved analytical issues through documentation, literature review, iterative testing, and team discussion.
- Interpreted ENO1-high glycolytic and hypoxic epithelial states, generated testable validation hypotheses, designed the figure framework, and drafted the manuscript.

TLS-related Immunogenetic Subtypes in Lung Adenocarcinoma

Nanchang, China

Lead Researcher and First Author

2025–2026

- Proposed and led a study testing whether TLS-related molecular features define clinically and immunologically distinct lung adenocarcinoma subtypes.
- Designed and implemented a framework integrating Mendelian randomization, survival modeling, immune profiling, TMB analysis, single-cell RNA-seq, and RT-qPCR validation.
- Synthesized the results into a coherent subtype framework, prepared figures, led manuscript writing, and collaborated with team members to troubleshoot analyses and align computational findings with experimental interpretation.

BMP-related mRNA Prognostic Modeling in Kidney Renal Clear Cell Carcinoma

Nanchang, China

*Lead Researcher and First Author**2025–2026*

- Initiated my formal training in cancer bioinformatics by independently reviewing the literature, identifying a clinically relevant research question, selecting the BMP-related topic, and designing the overall study framework.
- Self-learned R programming and implemented an end-to-end analytical pipeline using TCGA and an independent E-MTAB cohort, including differential expression analysis, correlation analysis, LASSO and Cox regression, survival modeling, external validation, TME/TMB analysis, and drug-sensitivity analysis.
- Iteratively diagnosed and resolved data-processing and modeling problems by consulting methodological literature, testing alternative strategies, and discussing unresolved issues with supervisors and peers.
- Led figure design, biological interpretation, and manuscript drafting, gaining experience in converting a clinical question into a reproducible computational study and a publishable first-author project.

RESEARCH COMPETENCIES

- **Question Formulation and Study Design:** literature synthesis, identification of research gaps, hypothesis development, selection of appropriate datasets and analytical strategies, and construction of project-specific study frameworks.
- **Self-directed Learning and Problem Solving:** independent acquisition of R-based methods, workflow customization, debugging, critical comparison of alternative analyses, and iterative refinement of study design.
- **Scientific Writing and Visualization:** manuscript drafting, figure planning and production, organization of analytical narratives, and translation of statistical results into biologically interpretable conclusions.
- **Collaboration and Communication:** clear discussion of methodological and biological problems, active exchange of feedback, coordination with computational and experimental collaborators, and constructive resolution of research challenges.

TECHNICAL SKILLS

- **Programming and Reproducibility:** R, Python, GitHub, Markdown, and LaTeX.
- **Transcriptomics and Multi-omics:** bulk RNA-seq, single-cell RNA-seq integration, spatial transcriptomics, WGCNA, pathway enrichment, pseudotime analysis, and ligand–receptor interaction inference.
- **Statistical and Cancer Bioinformatics Methods:** differential expression analysis, LASSO and Cox regression, survival modeling, machine-learning-based risk stratification, external validation, tumor microenvironment analysis, tumor mutational burden analysis, and drug-sensitivity analysis.
- **Databases and Platforms:** TCGA, GEO, E-MTAB, public single-cell RNA-seq datasets, and public spatial transcriptomics datasets.
- **Experimental Collaboration:** RT-qPCR-based gene-expression validation and interpretation of experimental results in collaboration with laboratory researchers.

CLINICAL TRAINING

- Completed systematic coursework in physiology, pathology, pharmacology, internal medicine, surgery, pediatrics, obstetrics and gynecology, and clinical reasoning.
- Completed clinical rotations across multiple departments, developing a clinically grounded perspective for defining research questions and interpreting cancer multi-omics findings, prognostic models, and patient-stratification frameworks.

LANGUAGES

Mandarin Chinese (native); English (academic reading and scientific writing).